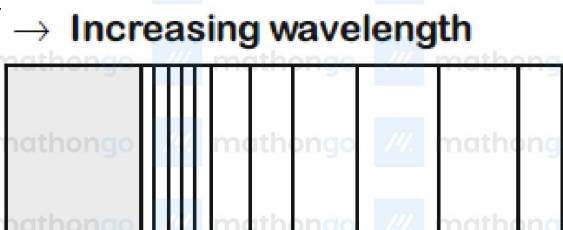


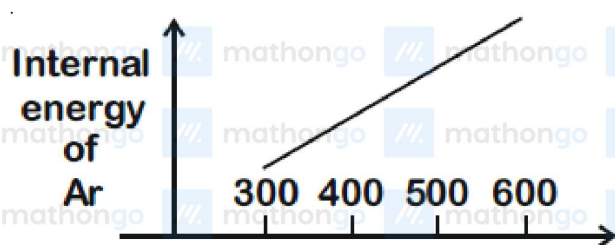
Q1 JEE Main 2020 - 2 September (Morning)

The figure that is not a direct manifestation of the quantum nature of atoms is



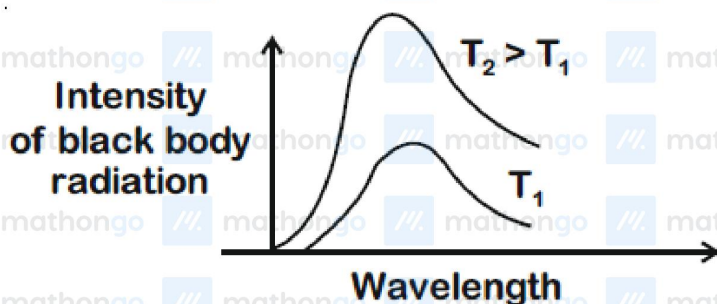
Absorption spectrum

(A)



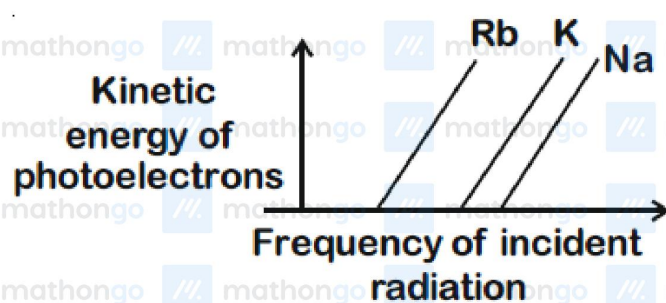
Temperature (K)

(B)



Wavelength

(C)



(D)

Atomic Structure

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

Q2 JEE Main 2020 - 2 September (Evening)

The number of subshells associated with $n = 4$ and $m = -2$ quantum numbers is

- (A) 2
- (B) 8
- (C) 4
- (D) 16

Q3 JEE Main 2020 - 2 September (Evening)

The work function of sodium metal is

$4.41 \times 10^{-19} \text{ J}$. If photons of wavelength 300 nm

are incident on the metal, the kinetic energy of the ejected electrons will be ($h = 6.63 \times 10^{-34} \text{ Js}$; $c = 3 \times 10^8 \text{ m/s}$) _____ $\times 10^{-21} \text{ J}$

Q4 JEE Main 2020 - 3 September (Evening)

Consider the hypothetical situation where the azimuthal quantum number, l , takes value $0, 1, 2, \dots, n-1$, where n is the principal quantum number. Then, the element with atomic number

- (A) 9 is the first alkali metal
- (B) 6 has a $2p$ -valence subshell
- (C) 8 is the first noble gas
- (D) 13 has a half-filled valence subshell

Q5 JEE Main 2020 - 4 September (Morning)

The region in the electromagnetic spectrum where the Balmer series lines appear is

- (A) Microwave
- (B) Ultraviolet
- (C) Visible
- (D) Infrared

Q6 JEE Main 2020 - 4 September (Evening)

The shortest wavelength of H atom in the Lyman series is λ_1 . The longest wavelength in the Balmer series of He^+ is

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Atomic Structure

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

(A) $\frac{5\lambda_1}{9}$

(B) $\frac{36\lambda_1}{5}$

(C) $\frac{27\lambda_1}{5}$

(D) $\frac{9\lambda_1}{5}$

Q7 JEE Main 2020 - 5 September (Morning)

In the sixth period, the orbitals that are filled are

(A) 6s, 4f, 5d, 6p

(B) 6s, 5d, 5f, 6p

(C) 6s, 6p, 6d, 6f

(D) 6s, 5f, 6d, 6p

Q8 JEE Main 2020 - 5 September (Evening)

The correct statement about probability density (except at infinite distance from nucleus) is

(A) It can never be zero for 2s orbital

(B) It can be zero for 3p orbital

(C) It can be zero for 1s orbital

(D) It can be negative for 2p orbital

Q9 JEE Main 2020 - 7 January (Morning)

The number of orbitals associated with quantum number $n = 5$, $m_s = +\frac{1}{2}$

(A) 25

(B) 30

(C) 50

(D) 35

Q10 JEE Main 2020 - 7 January (Morning)

Amongst the following which is not a postulate of Dalton's atomic theory

(A) Matter is formed of indivisible atoms

- (B) When gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P.
- (C) During chemical reactions atoms remains conserved and only pass through rearrangement
- (D) Some atoms have same properties including atomic mass

Q11 JEE Main 2020 - 8 January (Morning)

For the Balmer series in the spectrum of for H -atom $\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

Select the correct options regarding this formula for Balmer series.

- (a) The integer n_1 is equal to $= 2$
- (b) Ionization energy of H atom can be calculated from wave number of these lines.
- (c) The lines of longest wavelength corresponds to $n_2 = 3$.
- (d) If λ decreases then spectrum lines will converse.
- (A) A, B
- (B) C, D
- (C) $A \& C \& B$
- (D) $A, B, C \& D$

Q12 JEE Main 2020 - 8 January (Evening)

The radius of the second Bohr orbit, in terms of the Bohr radius, a_0 , in Li^{2+} is

- (A) $\frac{3a_0}{4}$
- (B) $\frac{4a_0}{3}$
- (C) $\frac{a_0}{3}$
- (D) $\frac{16a_0}{9}$

Q13 JEE Main 2020 - 9 January (Morning)

The *de* Broglie wavelength of an electron in the 4th Bohr orbit is

Atomic Structure

JEE Main 2020 Chapterwise

Questions with Answer Keys

Chemistry

(A) $4\pi a_0$

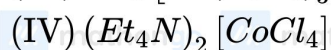
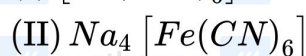
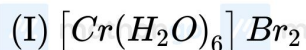
(B) $2\pi a_0$

(C) $8\pi a_0$

(D) $6\pi a_0$

Q14 JEE Main 2020 - 9 January (Evening)

The correct order of the spin-only magnetic moments of the following complexes is :



(A) $(III) > (I) > (IV) > (II)$

(B) $(III) > (I) > (II) > (IV)$

(C) $(I) > (IV) > (III) > (II)$

(D) $(II) \approx (I) > (IV) > (III)$

Atomic Structure

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Answer Key

Q1 (B)

Q2 (B)

Q3 (222)

Q4 (D)

Q5 (C)

Q6 (D)

Q7 (A)

Q8 (B)

Q9 (A)

Q10 (B)

Q11 (C)

Q12 (B)

Q13 (C)

Q14 (C)

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Q1

Explanation of variation of internal energy of Ar with temperature (Straight line and $U \propto T$) is not a direct manifestation of the quantum nature of atoms.

While explanation of absorption spectrum, nature of emission of radiation from hot bodies (black body radiation) and photoelectric effect are direct manifestation of the quantum nature of atoms.

Q2

2 subshells are associated with $n = 4$ and $m = -2$

Q3

w, work function of sodium metal

$$= 4.41 \times 10^{-19} \text{ J}$$

λ , wavelength of incident light = 300 nm

$$= 3 \times 10^{-7} \text{ m}$$

According to photoelectric effect

$$\frac{hc}{\lambda} = w + \text{KE}$$

$$\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3 \times 10^{-7}} = 4.41 \times 10^{-19} + \text{KE}$$

$$6.63 \times 10^{-19} = 4.41 \times 10^{-19} + \text{KE}$$

$$\text{KE} = 2.22 \times 10^{-19} \text{ J} = 222 \times 10^{-21} \text{ J}$$

Q4

Under hypothetical situation, the value of l is greater than n which varies from 0 to $n + 1$

n for $n = 1, l = 0, 1, 2$

$n = 2, l = 0, 1, 2, 3$

Elements follow the following electronic configuration

$1s^1 1d^1 2s^2 2p^2 2f^1$

Atomic number (Z) = 9

$1s^2 1p^6 1d^1$

Atomic number 6

$1s^2 1p^4$

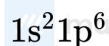
Atomic number 8

Atomic Structure

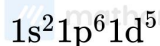
Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry



Atomic number 13



Here atomic number of first noble gas will be 18

Q5

In the hydrogen spectrum,
Balmer series lies in visible region.

Q6

Shortest wavelength $\rightarrow$ Max energy ($\infty \rightarrow 1$) (Lyman series)

$$\frac{1}{\lambda_1} = R_H (1)^2 \left[\frac{1}{1} - 0 \right]$$

$$\frac{1}{\lambda_1} = R_H \Rightarrow R_H = \frac{1}{\lambda_1}$$

For Balmer series,

$$\frac{1}{\lambda} = R_H (2)^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \Rightarrow \frac{1}{\lambda} = R_H (4) \left(\frac{9-4}{36} \right)$$

$$\frac{1}{\lambda} = \frac{5R_H}{9} \Rightarrow \lambda = \frac{9}{5R_H} = \frac{9\lambda_1}{5}$$

Q7

Filling of electrons in orbitals in any period takes place as:

$ns, (n-2)f$

$(n-1)d$

if possible

$\therefore$ for sixth period $n = 6$

orbitals that are filled are 6s, 4f, 5d and 6p

Q8

ϕ^2 (probability density) can be zero for 3p orbital other than infinite distance. It has one radial node.

Q9

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Atomic Structure

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Self-Explanatory

Q10

Self-Explanatory

Q11

Theory Based

Q12

$$r = \frac{a_0 n^2}{2}$$

$$\text{For } Li^{2+} \quad r = \frac{a_0 (2)^2}{3} = \frac{4a_0}{3}$$

Q13

$$2\pi r = n\lambda$$

$$2\pi \times \frac{n^2}{Z} a_0 = n\lambda$$

$$2\pi \times \frac{4^2}{1} a_0 = n\lambda$$

$$\lambda = 8\pi a_0$$

Q14

Atomic Structure

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

$$\text{As, } \mu_s = \sqrt{n(n+2)}$$

$$\text{Complex (I)} \rightarrow Cr^{2+} \Rightarrow \text{W.F.L} \Rightarrow t_{2g}^{1,1,1} e_g^{1,0}$$

$$\Rightarrow \mu_s = \sqrt{24} \text{ B.M}$$

$$\text{Complex (II)} \rightarrow Fe^{+2} \Rightarrow \text{S.F.L} \Rightarrow t_{2g}^{2,2,2} e_g^{0,0} \Rightarrow \mu_s = 0$$

$$\text{Complex (III)} \rightarrow Fe^{+3} \Rightarrow \text{S.F.L} \Rightarrow t_{2g}^{2,2,1} e_g^{0,0}$$

$$\Rightarrow \mu_s = \sqrt{3} \text{ B.M.}$$

$$\text{Complex (IV)} \rightarrow Co^{+2} \Rightarrow \text{W.F.L} \rightarrow e_g^{2,2} t_g^{1,1,1}$$

$$\Rightarrow \mu_s = \sqrt{15} \text{ B.M.}$$